Proposal for a collaborative project in response to the special call on

Digital Health in Developing Countries

by Else Kröner Fresenius Foundation

A Project description

1. General information

Project title

Telehealth Services for Myanmar (Telekyanmar Project)

Applicant organisation (grant recipient)

Goethe-Universität Frankfurt
Theodor-W.-Adorno-Platz 1
60323 Frankfurt am Main
Germany

https://www.uni-frankfurt.de

Authorised representative

Dr. Sabine Monz
Bereichsleiterin Research Support
Goethe-Universität Frankfurt
Theodor-W.-Adorno-Platz 1
60629 Frankfurt am Main
Tel.: 0049 69 798 17403
E-Mail: monz@pvw.uni-frankfurt.de

Project manager

Dr. Ulrich Kuch
Abteilung Tropenmedizin und Global Health
Institut für Arbeits-, Sozial- und Umweltmedizin (Direktor: Prof. Dr. Dr. med. David Groneberg)
Goethe-Universität Frankfurt
Theodor-Stern-Kai 7
60590 Frankfurt am Main
Germany
Tel.: 0049 170 5952010
E-Mail: kuch@med.uni-frankfurt.de

Local partner organisation

Organisation Name: Telekyanmar
Registration No.: NGO - 36 / 23 / 12
Country: Myanmar
Program manager: Dr. Laxus Diego
Tel.: 0082 10 4200-6618
E-mail: laxus@telekyanmar.org

Address:

Room - 1512, Building 1
Pilleo Twin Park
Jungdong 1160-2
Bucheon-si
Gyeonggi-do
South Korea

Brief description of all organisations actively involved in the project

Goethe University Frankfurt

Goethe University (GU) was founded in 1914 as a unique “citizens’ university” in Frankfurt. Home to over 41,000 students, including about 8,000 international students from 150 countries, it is the third largest university in Germany. GU maintains partnerships and collaborative projects with 700 universities all over the world. In keeping with its mission to support civic society, GU was one of the first higher education institutions in Germany to establish a dedicated academic pathway programme for refugees. The Faculty of Medicine of GU and University Hospital Frankfurt (UKF) with over 1300 physicians and scientists and about 2700 students in more than 50 specialist clinics, clinical, and research institutes ensure the integration of teaching, research, and patient care. We combine the special responsibility of training future generations of healthcare professionals with the opportunities created by integrating evidence-based scientific findings directly into patient care.

Telekyanmar Telemedicine Organization

TeleKyanmar was established as a telemedicine program in 2021 by the Ministry of Health (MOH) of the National Unity Government of Myanmar (NUG) with the goal of providing essential medical services to people who faced challenges accessing healthcare during the current multidimensional crisis in Myanmar. Telekyanmar is now a registered NGO in Myanmar. Its mission is ensuring accessible healthcare for all, adapting to the evolving needs and circumstances in Myanmar in order to continue serving those who rely on telemedicine services for their well-being, and to build a sustainable model for future health service delivery across all of Myanmar. Telekyanmar, being an already established and operational telemedicine program, has achieved significant success in improving healthcare access for remote communities in Myanmar since June 2021. It has successfully provided more than 180,000 of remote consultations, facilitated timely diagnosis, and strengthened local healthcare capacity through training programs. Collaborating with ethnic health organisations in Myanmar, the program has overcome geographical, cultural, religious and linguistic barriers, ensuring that quality healthcare reaches underserved areas. With a proven track record of positive health outcomes and strong community engagement, Telekyanmar continues to expand its reach, demonstrating the potential of telemedicine to transform healthcare delivery across Myanmar’s diverse regions.

Karen Department of Health and Welfare

The Karen Department of Health and Welfare (KDHW) is an ethnic health organization (EHO) dedicated to serving the health and welfare needs of the Karen population in Myanmar. Established in 1956, KDHW provides comprehensive healthcare services, including primary and secondary care, through an extensive network of Village Tract Health Centers and secondary care centers. Its mission is to ensure that all individuals in the areas of Myanmar that are administered by the Karen National Union (KNU) areas have access to quality, affordable, and equitable health and welfare services. KDHW focuses on community-based healthcare, addressing common diseases, maternal and child health, emergency obstetric care, and health education.

Health Department, Kachin Political Interim Coordination Team

The Health Department of the Kachin Political Interim Coordination Team (KPICT) operates as part of a broader effort to represent and coordinate the political aspirations of the Kachin people in Myanmar. Formed in March 2021, KPICT aims to end authoritarianism and establish a genuine Federal Democratic Union in Myanmar. The Health Department of KPICT focuses on providing essential healthcare services to the population in Kachin State of northern Myanmar, particularly in areas affected by conflict and displacement. These initiatives include distributing medical supplies, managing emergency health situations, and collaborating with both domestic and international organizations to improve health outcomes.

Health Department, Karenni State Interim Executive Council

The Health Department of the Karenni State Interim Executive Council (IEC) is a key component of the provisional government established in the Karenni State of Myanmar in June 2023. The Karenni State IEC was created to unify various Karenni resistance groups in their defence against the military junta of Myanmar and to provide stable governance in resistance-held areas. The Health Department of the Karenni State IEC is responsible for delivering essential healthcare services, including primary care, maternal and child health, and emergency medical responses. Its goal is to ensure that all residents of Karenni State have access to necessary health services, thereby improving overall health and well-being in the region.

Ministry of Health, National Unity Government of the Republic of Union of Myanmar

The interim National Unity Government of the Republic of the Union of Myanmar (NUG) is the only and legitimate Government of the Republic of the Union of Myanmar. After the illegal military coup of February 2021, the NUG was formed by the Committee Representing Pyidaungsu Hluttaw (CRPH) in discussion with the National Unity Consultative Council (NUCC) of Myanmar. The CRPH consists of members of the parliament of Myanmar who had been elected by the people of Myanmar in the last free, fair and democratic election held in this country in November 2020. The NUG implements the goal, objective and steps of the political road map for Myanmar as prescribed in Part I of the Federal Democracy Charter in discussion with the NUCC, and is accountable to the CRPH. The Ministry of Health (MOH) of the NUG is responsible for healthcare delivery and public health at the federal/union level in Myanmar and for the transformation of health policies and the health system towards federalism according to the Federal Democracy Charter. Its goal is to ensure that the people of Myanmar have access to the health services they need during the interim period of the ongoing anti-dictatorship movement by the people, which aims for the return of the power to the people, genuine freedom, federal democracy, equality, rule of law, and sustainable peace. The MOH follows its Strategic Healthcare Plan, the two cornerstones of which are Universal Health Coverage (UHC) and Federal Democratic Principles.

Project duration in months

The proposed project duration is 36 months.

Overview of the financing

The total direct implementation costs of the proposed project will be € 500,000 €. This amount, i.e., 100% of the specific project cost, is applied for from the EKFS.

Of this total amount, approximately 32% will be spent on human resources, 9% on travel, 13% on project activities, 22% on consumables including medicines, and 25% on other costs.

A detailed budget in Euros of the funds requested from the EKFS is provided separately in the Excel file (part "E Budget").

The tele-touring project proposed here is a part of the Telekyanmar telemedicine program and will be implemented in close collaboration with three ethic health organizations (EHOs) in Myanmar and the Institute of Occupational, Social and Environmental Medicine of Goethe University Frankfurt. Each of these five organizations will make their existing facilities available to the project as needed as a part of their own contributions.

Activities of health promotion through Facebook, producing health promotion materials, salary for the senior program management teams of Telekyanmar and the three EHOs as well as the salaries of other healthcare providers not listed in the proposed financial plan of this project are indirectly contributed by other donor projects and the budgets of the EHOs, respectively.

The project partners plan to have a mid-term audit (after 18 months from project start) and intend to task a regional branch of the auditing organisation BDO with this audit.

2. Project description

Context and significance of the project

Healthcare access remains a significant challenge in Myanmar, particularly for rural and remote communities. Limited infrastructure, a shortage of healthcare professionals, and geographical barriers hinder timely access to essential medical services. In light of these challenges, telemedicine presents a transformative opportunity to bridge the healthcare gap, leveraging digital technology to connect patients with healthcare providers regardless of location.

This project proposal aims to contribute to establishing a comprehensive telemedicine program in Myanmar, focusing on delivering quality healthcare services to underserved areas, particularly to those that are administered by ethnic minority organizations and have long been affected by protracted armed conflicts with successive military dictatorships in Myanmar. By integrating telemedicine platforms with existing healthcare resources on the ground, we aim to provide real-time consultations, remote diagnostics, and continuous medical education to healthcare professionals in remote locations. The proposed project will utilise mobile applications with low bandwidth that are accessible on various devices, ensuring flexibility and user-friendliness for both patients and healthcare providers.

This telemedicine project, implemented in Myanmar by the Telekyanmar Team, will collaborate closely with ethnic health organisations (EHOs) in the Kachin, Karen, and Karenni States of Myanmar to ensure culturally sensitive and effective healthcare delivery. The Telekyanmar Team will establish partnerships with local health organisations to understand the unique healthcare needs and challenges of these communities. This collaboration will involve co-designing telemedicine services, integrating traditional healthcare practices, and aligning with local health priorities.

Ethnic health organisations in the Kachin, Karen, and Karenni States will serve as the primary providers of local healthcare services, directly linking with the Telekyanmar Team to enhance service delivery. They will utilise the telemedicine platform to connect patients with remote medical specialists, facilitate real-time consultations, and provide diagnostic support. Local health workers will conduct initial patient assessments, manage follow-ups, and ensure continuity of care, while leveraging the Telekyanmar Team's technical expertise and resources. This collaboration will create a seamless integration of local and remote healthcare services, improving access to specialised care, enhancing treatment outcomes, and strengthening the overall health system in these communities.

The Telekyanmar Team will provide data security training and capacity-building support to local healthcare workers in telemedicine technologies, ensuring they are equipped to deliver remote consultations, diagnostics, and follow-up care effectively. The project will deploy multilingual telemedicine platforms to bridge language barriers, enabling communication in local languages like Kachin, Karen, and Karenni. In addition, the Telekyanmar team will provide Mental Health and Psychosocial Support (MHPSS) training to health care workers on the ground to equip them with the capability to provide health awareness in the community, psychosocial first aid, and access and referral to needed psychiatrist care which is available at the Telekyanmar Team and selected health facilities of the participating EHOs.

To build trust and encourage community participation, the Telekyanmar Team will work alongside EHOs to conduct awareness campaigns and community engagement activities. Regular feedback sessions will be held with local partners to assess the program's impact, identify challenges, and adapt strategies accordingly.

By implementing this telemedicine project, we aim to reduce healthcare disparities, improve health outcomes, and build a more resilient healthcare system in Myanmar, paving the way for a healthier and more equitable future.

Preliminary work

Telekyanmar as a telemedicine program was established in 2021 by the Ministry of Health (MOH) of the National Unity Government of Myanmar (NUG) to address the critical healthcare challenges faced by individuals during the ongoing multidimensional crisis of this country. Telekyanmar has since been registered as an NGO in Myanmar. Its mission is built on three foundational pillars: service provision, health promotion, and research. These pillars ensure accessible healthcare for all, adapt to the evolving needs in Myanmar, and build a sustainable model for future health service delivery across the country.

Initially responding to the urgent healthcare needs arising from the COVID-19 pandemic, the Telekyanmar program has expanded after the 2021 military coup in Myanmar because the latter caused mass population displacement to remote jungle and mountain areas, a collapse of conventional facility-based health services in areas controlled by the military, and escalating armed conflict across the country. By 2022, Telekyanmar had developed primary mental health clinics and enhanced health promotion initiatives, utilising various regional languages. Continuous activities throughout 2023 included partnerships with healthcare teams on the ground to facilitate provider-to-provider referrals, technical support for electronic medical records (EMR), and mental health training. In 2024, Telekyanmar launched the ‘Mental Health Wellbeing for All’ project, which features a Zero Suicide Campaign, a suicide prevention hotline, and psychiatric referral assistance.

The Telekyanmar program is led by a skilled support team, including a program manager, finance department, HR department, organisational development specialists, M&E personnel, IT staff, and clinic managers, all coordinated by a program committee. The quality assurance department oversees the overall quality of telemedicine activities. Telekyanmar has a robust capacity development program that conducts training sessions, continuing medical education (CME), and well-being activities for its volunteers. Over the past three years, Telekyanmar has

significantly expanded, now comprising over 307 volunteer clinicians providing telehealth services across primary and secondary care in 25 specialties.

The innovative health promotion efforts of Telekyanmar include active social media engagement, with a website of the program (https://telekyanmar.org), a Facebook page that has 470,000 followers (https://facebook.com/telehealthmm), and a YouTube channel (https://www.youtube.com/@telehealthmm). By the end of August 2024, Telekyanmar had facilitated a total of 190,000 teleconsultations for 62,058 patients, reaching 322 out of 330 townships in Myanmar, with 67% of beneficiaries being female. Furthermore, the 2023 satisfaction surveys of Telekyanmar showed that more than 86% of providers and patients expressed satisfaction with the services received and provided by the program.

The Telekyanmar telemedicine program utilises a dual approach for consultations. For health education and awareness, it leverages Facebook to reach a wide audience, sharing informative content, health tips, and updates in local languages to engage communities. For consultations and treatment, the project employs a bespoke communication channel designed specifically for secure, real-time interactions between patients and healthcare providers. This platform facilitates direct video consultations, messaging, and remote monitoring, ensuring confidentiality and efficiency. By combining a widely accessible social media platform with a tailored, secure communication tool, the project ensures comprehensive and accessible healthcare delivery across diverse communities.

Telekyanmar collaborates closely with hospitals and primary healthcare facilities to provide emergency health services, specialist consultations, and capacity building in mental health and psychosocial support. Our technology support and health promotion initiatives have addressed language barriers, ensuring broader access to healthcare. As the demand for telemedicine services remains high in Myanmar, the program fulfils a vital need for countless patients nationwide.

3. Project objectives and indicators

The goal of this telemedicine project is to improve health care access and quality in conflict-affected areas of Myanmar through telemedicine and capacity-building initiatives.

The project has three main objectives:

(1) Promote accessibility of patients to real-time specialist consultations, remote diagnostics, and follow-up care.

(2) Train providers in the use of telemedicine technologies to ensure effective service delivery.

(3) Train healthcare providers in the delivery of MHPSS health promotion activities and linkage to mental health services at Telekyanmar telemedicine

Outcome (outcome indicators):

- Enhanced healthcare delivery and provider capabilities in ethnic health organizations (EHO) in conflict-affected areas of Myanmar (a total of 9,000 consultations and 1,800 specialist consultations conducted in the service areas of the three participating EHO through mobile clinics by the end of 2027)

Outputs (output indicators):

- Tele-touring telemedicine services implemented (a total of 360 tele-touring visits conducted in the service areas of the three participating EHO by the end of 2027)
- MHPSS and data security training delivered (baseline MHPSS training conducted in the service areas of all three participating EHO; a total of 60 healthcare providers attended MHPSS training; a total of three data security training sessions conducted in the service areas of all three participating EHO; a total of 60 participants attended data security training)

- MHPSS coordination workshop conducted (a total of two MHPSS coordination workshops conducted in 2026 and 2027 in the service areas of all three participating EHO; a total of 180 MHPSS related health promotion sessions conducted at the time of mobile clinics)
- Health promotion contents developed and disseminated in major regional/local languages (a total of 36 health promotion materials developed by the end of project year 3, in three major regional/local languages in addition to Burmese)

The goal, objectives, outcome and outputs as well as the indicators and activities are presented in detail in the separate Excel logframe (part D of this application).

The proposed telemedicine project will adopt a multi-faceted strategy to ensure effective and sustainable healthcare delivery. It will collaborate with ethnic health organisations in the Kachin, Karen, and Karenni States of Myanmar to integrate local healthcare services with Telekyanmar's remote consultation platform, ensuring cultural sensitivity, relevance and trust. To overcome internet interruptions and power outages, the project will use offline-capable apps, SMS-based services, solar-powered devices, and portable battery packs. It will leverage Facebook for health education and awareness while using a secure, bespoke communication channel for confidential consultations and treatment. Regular feedback from local partners and communities will guide program adaptations, ensuring services are responsive to evolving needs. By combining technology with community engagement and resilience measures, the project aims to provide sustainable, accessible, and quality healthcare to underserved populations.

To coordinate with ethnic health organisations, the project will hold regular virtual and in-person meetings to discuss progress, challenges, and strategies. Coordination meetings will be scheduled every two weeks for all parties to ensure effective participation at the beginning of the project, and will later be continued in monthly intervals. Resource relocation will be managed through a project implementation body where ethnic health leaders join, with timely updates provided to local organisations. Collaborative planning sessions will ensure that resource distribution aligns with local needs and priorities. Additionally, dedicated liaison officers will facilitate communication and address any issues that arise, ensuring smooth and responsive collaboration.

4. Beneficiaries

The primary beneficiaries of this project are patients and their families, as well as healthcare professionals and volunteers in Myanmar of the Telekyanmar program and EHOs. Secondary beneficiaries include the public health systems in previously underserved regions of Myanmar.

The proposed direct beneficiaries among patients will be 10,800 persons who receive medical services. Among these, 1,800 persons will receive specialist medical services.

Among the providers, 60 healthcare providers from EHOs will receive online training regarding MHPSS topics and will participate in the MHPSS coordination workshop on an annual basis. In addition, 60 providers from EHOs will receive online data security training on an annual basis.

The direct beneficiaries among healthcare professionals will be 1 Deputy Project Manager, 1 Finance Coordinator and 1 M&E Coordinator to manage and contribute Telekyanmar telemedicine, and healthcare professionals from 3 EHOs. From the EHOs, there will be 3 Project Coordinators and 6 on-ground healthcare providers to coordinate and implement the activities. From Telekyanmar organisation, 1 Human Resource Officer, 1 Data Governance Officer, 1 Medical Doctor and 50 volunteers will support and implement the activities.

The project will ensure equitable distribution of benefits between the Telekyanmar program and the three EHOs which provide access to a range of populations including those affected by conflict, displaced individuals, and ethnic minorities, all of whom are registered as patients.

Currently, 60.8% (188 out of 309) of the registered and active healthcare providers under Telekyanmar are women, reflecting the program's commitment to gender sensitivity. Additionally, health education materials will be developed and distributed in three regional languages promoting accessibility for diverse communities. To further promote inclusivity, the gender focal point of the Telekyanmar program will offer regular gender equality and social inclusion sharing sessions to the project staff including providers from EHOs.

5. Perspective / sustainability

While the Telekyanmar program currently benefits from two funding sources, additional funding is crucial for expanding service provision and ensuring long-term sustainability. New, project-specific funding will enable the expansion of telemedicine services to more remote areas, increase the number of healthcare providers trained in telemedicine, and upgrade the technology infrastructure to enhance platform reliability and accessibility. It will also support the development of localised health content, translation services, and community outreach initiatives, ensuring that services are culturally appropriate and widely accepted. Furthermore, additional funding will help establish a sustainable financial model through capacity building, operational scaling, and the integration of innovative financing mechanisms, securing the program's future viability.

When the political situation in Myanmar settles down, the Telekyanmar telemedicine program will continue to run as a government funded program financed from the budget of the Federal Ministry of Health of Myanmar.

6. Risks

The economic, public health and security situation in Myanmar has been declining since the 2021 military coup. The military junta has very limited capacity to provide public services in those areas that are still under its control. Civilian administration structures by the interim National Unity Government and allied ethnic organizations are in regionally variable stages of establishment and operation. Thus, the proposed project has been designed with several calculated risks in mind, and the following mitigation plan in place to address these:

(1) Internet Disruptions**:** Frequent internet interruptions could limit access to telemedicine services.

Mitigation: Use offline-capable mobile apps, SMS-based services, and deploy local servers to maintain functionality during outages.

(2) Electricity Outages**:** Regular power outages may disrupt consultations and device use.

Mitigation: Provide solar-powered devices, portable battery packs, and power banks to healthcare workers and patients.

(3) Security Risks: Military surveillance and conflict pose risks to the safety of health workers and data privacy.

Mitigation: Use encrypted communication channels, store data securely, and train staff in safety protocols.

(4) Security and Transportation Challenges: Delays in opening mobile clinics due to security threats and transportation difficulties.

Mitigation: Establish regular security assessment, alternative mode of transportation and contingency plan for operations.

(5) Community Acceptance: Cultural differences and lack of trust in digital health solutions could hinder uptake.

Mitigation: Collaborate with ethnic health organisations to build trust, offer services in local languages, and conduct community awareness campaigns.

7. Signatures of the applicants

JOHANN WOLFGANG GOETHE-UNIVERSITÄT
Der Präsident
Theodor-W.-Adorno-Platz 1
60323 Frankfurt am Main

i.A. Sabine Monz

Dr. Sabine Monz
Authorised representative

Project management
Dr. Ulrich Kuch
Institut für Arbeits-, Sozial- und Umweltmedizin
Goethe-Universität
Theodor-Stern-Kai 7
60590 Frankfurt am Main